

Total No. of Questions : 4]

SEAT No. :

P8559

[Total No. of Pages : 2

Oct-22/TE/Insem-529

T.E. (Computer Engineering)

INTERNET OF THINGS AND EMBEDDED SYSTEMS

(2019 Pattern) (310245(A)) (Semester - I) (Elective - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Attempt Q1 or Q2, Q3 or Q4.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

Q1) a) What is an embedded system? What are the characteristics of an embedded system? [5]

b) Introduce any embedded processor in brief. Explain its architecture. [5]

c) Differentiate between General Computer and embedded devices. [5]

OR

Q2) a) Define SOC. Illustrate SOC types and its examples. [5]

b) Illustrate the different components of Microcontroller. [5]

c) Explain the concept of RTOS. [5]

Q3) a) Explain the concept of 'Things' in IoT with suitable examples. [5]

b) Enlist and Demonstrate societal benefits of IoT. [5]

c) What are the challenges in implementing IoT Applications? [5]

OR

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- Q4)** a) Enlist IoT deployment levels and explain IoT level 2 with suitable application. [5]
- b) Define IoT and explain its importance in the real-world problem solving. [5]
- c) Illustrate the physical design of IoT with suitable example. [5]



Total No. of Questions : 4]

SEAT No. :

P5029

[Total No. of Pages : 1

[6187]-429

T.E. (Computer Engineering) (Insem)
INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (310245(A)) (Elective - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) What is an embedded system? What are the characteristics of an embedded system? [5]
b) Introduce any embedded processor in brief. Explain its architecture. [5]
c) What is a real time system? Describe the types of real time tasks. [5]

OR

- Q2)** a) Draw and elaborate the general model of an embedded system, what are the different applications of an embedded system. [5]
b) Illustrate the different components of Microcontroller. [5]
c) Define SOC. Explain it with suitable examples of it. [5]

- Q3)** a) Explain the concept of 'Things' in IoT with suitable examples of IoT devices. [5]
b) Illustrate the need of analog to digital signal conversion and vice versa. Explain it with suitable application. [5]
c) What are the challenges in implementing IoT applications? [5]

OR

- Q4)** a) Enlist IoT deployment levels and explain IoT level 4 with suitable application. [5]
b) Define IoT and illustrate the use of Raspberry Pi as an IoT Device with suitable application. [5]
c) Illustrate the Logical design of IoT with suitable example. [5]



Total No. of Questions : 4]

SEAT No. :

PC29

[6360]-29

[Total No. of Pages : 1

T.E. (Computer Engineering) (Insem)
INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester-I) (Elective - I) (310245 A)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) Figures to the right indicate full marks.*

- Q1)** a) What is System - On - Chip? Discuss the advantages of it. [5]
b) Discuss a hard real - time system with a suitable example. [5]
c) Discuss the one of given embedded systems and its components ATM Machine/ Digital Camera. [5]

OR

- Q2)** a) What are the desirable features of an embedded systems? [5]
b) Explain soft and hard real - time tasks with suitable examples. [5]
c) Discuss the components of the microcontrollers with a diagram. [5]

- Q3)** a) Explain IOT level - 3 with suitable Application and diagram. [5]
b) Discuss that IOT systems have to be self - adaptive and self - configuring. [5]
c) What are the underlying characteristics of big data? [5]

OR

- Q4)** a) Enlist and explain protocols used in IoT. [5]
b) Explain the Publish - Subscribe communication model with a diagram. [5]
c) Draw a generic block diagram of IoT and discuss the working of each block. [5]



Total No. of Questions : 4]

SEAT No. :

PE28

[Total No. of Pages : 1

[6579] 329

**T.E. (Computer Engineering) (Computer Science) (Insem)
INTERNET OF THINGS AND EMBEDDED SYSTEMS
(2019 Pattern) (Semester - I) (310245 (A) (Elective-I)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Figures to the right indicate full marks.

- Q1)** a) List and explain the characteristics of the Embedded System. [5]
b) What is System-On-Chip? Explain its features. [5]
c) Discuss the types of microcontrollers [5]

OR

- Q2)** a) Describe the societal benefits of IoT with the help of suitable applications. [5]
b) Compare & contrast between microprocessor & microcontroller. [5]
c) Discuss the components of the embedded system with a diagram [5]

- Q3)** a) Write a definition of IoT and Characteristics of IoT [5]
b) How Cloud computing technology play role in IoT application [5]
c) Discuss the working of the request-response communication model with example. [5]

OR

- Q4)** a) What are the building blocks of the IoT system? [5]
b) Compare REST and Websocket communication model. [5]
c) Enlist IoT deployment levels and explain IoT level 4 with suitable Application. [5]

